

IDIRIMUNI SANUJA MENATH

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PROFESSIONAL SUMMARY

Software Engineering graduate and Information Systems scholar with hands-on experience building full-stack web applications, RESTful APIs, and scalable backend services. Proficient in modern frameworks including React, FastAPI, Python, and TypeScript, alongside relational and NoSQL databases. Demonstrated success in managing CI/CD pipelines, containerized environments, and system architecture design. Seeking a Software Engineering Internship to leverage strong technical problem-solving and software design skills in delivering high-impact software solutions.

TECHNICAL SKILLS

Languages:	Python, TypeScript, JavaScript, Java, C, C++, Rust, SQL, HTML/CSS
Frameworks & Libraries:	React.js, FastAPI, Spring Boot, Flutter, Tauri, Tailwind CSS
Databases & Cloud:	PostgreSQL, MongoDB, MySQL, Supabase, ChromaDB
DevOps & Tools:	Git, GitHub Actions, Podman, Docker, Vite, WebSockets, REST APIs
Core Competencies:	Full-Stack Architecture, CI/CD Pipeline Management, Edge AI / RAG, System Design, Agile/Scrum

TECHNICAL PROJECTS

Lumina LMS – AI-Powered Lecture Planning & Assessment System

Full-Stack & AI Systems Architect

Technologies: Python, FastAPI, React.js, TypeScript, LangGraph, Ollama, ChromaDB, MongoDB, WebSockets

- Architected a full-stack, microservices-based Learning Management System leveraging Edge AI to automate academic lecture planning and assessment workflows with zero-cost local inference.
- Engineered an AI Orchestration Microservice using **LangGraph** and **Ollama**, implementing a Generator-Critic pipeline for MCQ generation and a RAG pipeline utilizing **ChromaDB** and vector embeddings.
- Built real-time communication features using **FastAPI WebSockets** for instant notifications and live assessment updates across student, lecturer, and admin portals.
- Integrated a machine learning score prediction model ($R^2 = 0.84$) into student and admin predictive dashboards to proactively identify at-risk learners based on study habits and engagement metrics.

MoneyFlow – Cross-Platform Financial Ecosystem

Lead Software Architect

Technologies: Tauri 2.x (Rust), React 19, TypeScript, Expo 54 (React Native), Supabase (PostgreSQL), Vite, GitHub Actions

- Engineered a full-stack, cross-platform financial application using a monorepo architecture, enabling real-time cloud synchronization across Windows, macOS, Linux, Android, and iOS.
- Integrated **Supabase Auth** with Row Level Security (RLS) to enforce multi-tenant data privacy and secure real-time data sync across desktop (**Tauri/Rust**) and mobile (**Expo/React Native**) platforms.
- Architected a data migration bridge enabling local **SQLite** database synchronization into a centralized cloud **PostgreSQL** database without data loss.
- Implemented continuous integration (CI/CD) pipelines via **GitHub Actions** to automate builds, linting, type-checking, and native package distribution.

EDUCATION & CERTIFICATIONS

BSc (Hons) in Software Engineering | Cardiff Metropolitan University (UK)

Graduated with Second Class Honors (Upper Division)

BSc (Hons) in Information Systems (*Undergraduate*) | Sabaragamuwa University of Sri Lanka

Diploma in English & Information Technology | AOG Campus, Sri Lanka

G.C.E. Advanced Level & Ordinary Level | Holy Cross College, Kalutara

- G.C.E. Advanced Level (2022): Passed with 1 Distinction (B), 2 Credit passes (C)
- G.C.E. Ordinary Level (2019): Passed with 8 High Distinctions (A), 1 Distinction (B)